We will focus on the second strategy, the pruning criterion: in particular, we can provide a method that, given a parameter `ϵ`, called the *approximation error*, with `0.0 ≤ ϵ ≤ 0.5`, guarantees that in an approximated `n`-nearest neighbor search the `n`-th nearest neighbor returned will be within a factor `(1+ϵ)` from the true `n`-th nearest neighbor.

To explain this, let's restrict[15] to the case where `n==1`, just plain NN search; assume the approximated NN-search method, called on a point `P`, returns a point `Q`, while the real nearest neighbor of `P` would be another point `N≠Q`. Then, if the distance between `P` and its *true* nearest neighbor `N` is `d`, the approximated search distance between `P` and `Q` will be at most `(1+ϵ)*d`.

Guaranteeing this condition is easy: when we prune a branch, instead of checking if the distance between the target point and the branch's bounding envelope is smaller than current NN distance, we prune unless it is smaller than `1/(1+ϵ)` times the NN's distance.

If we denote with `Z` the query point, with `C` current nearest neighbor, and with `A` the closest point to `Z` in the branch pruned (refer to figure 10.21), we have, in fact:

$$d(Z,A) \geq \frac{1}{1+\epsilon} \cdot d(Z,C) \Rightarrow \frac{d(Z,C)}{d(Z,A)} \leq 1+\epsilon$$

So, if the distance of the closest point in the branch is higher than the reciprocal of `(1+ϵ)`, we are guaranteed that the